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MicroPPE-Guard: an efficient model for accurate personal protective equipment

Xuan Le, Phu Pham, Kien-T. Le, Luan Bui, Bao Huynh *

Faculty of Information Technology, HUTECH University, Ho Chi Minh City 700000, Viet Nam

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ABSTRACT

Compliance with safety regulations is essential for mitigating risks to workers across various sectors, including industry, construction, and public environments. Automated personal protective equipment (PPE) detection has become an important computer vision task for real-time safety monitoring, while also helping to reduce reliance on manual inspection. This research introduces a new compact PPE dataset comprising 25,000 high-resolution images spanning seven categories: safety boots, safety glasses, gloves, helmets, heads, persons, and reflective vests. In addition, we propose MicroPPE-Guard, a novel, effective, lightweight, and robust object detection architecture explicitly designed for small-scale PPE recognition. MicroPPE-Guard incorporates several architectural innovations, including an enhanced feature pyramid network, Squeeze-and-Excitation–based feature fusion, a decoupled detection head, and a Temporal–Spatial Compliance Module (TSCM) that enhances temporal stability and PPE–person compliance consistency for small and occluded PPE items. The proposed model achieves up to 92.4% mAP_{50} on the in-house 25K dataset while maintaining real-time inference performance. To further assess generic small-object transferability, the architecture was evaluated on UAVOD-10, a non-PPE UAV remote-sensing benchmark containing 844 samples, showing competitive performance under small-object and weak-feature conditions. The model was further validated on NewPPE25, an external PPE dataset, confirming cross-domain robustness across heterogeneous data sources. The model is designed for integration into real-time monitoring systems, in which its lightweight architecture supports the timely identification of potential PPE violations. Overall, the experimental results demonstrate that the proposed approach significantly enhances detection performance, paving the way for more effective safety protocol enforcement and improved worker protection across diverse operational environments.

1. Introduction

For decades, workplace safety in construction has remained a persistent challenge across both industrialized and developing countries. Despite considerable advances in safety protocols, inspection procedures, occupational risk control, and regulatory frameworks, the construction industry remains one of the most dangerous sectors worldwide. According to the International Labour Organization (ILO), as of 2023, nearly 3 million people die each year from work-related accidents or diseases[1], and the construction sector accounts for a substantial share of these fatalities [1]. A similar trend has been reported in the United States, where the U.S. Bureau of Labor Statistics (BLS) indicates that construction accounts for approximately one in every five worker deaths. Moreover, the Center for Construction Research and Training (CPWR)[2] reported a fatal injury rate of 9.5 per 100,000 full-time-equivalent (FTE) workers in construction, almost three times higher than the all-industry average of 3.2 per 100,000 FTE. Nonfatal injuries also remain a major concern, with over 200,000 cases reported in 2023, resulting in significant lost workdays and long-term disabilities[3].

* Corresponding author.
E-mail addresses: thanhxuan8054@gmail.com (X. Le), pta.phu@hutech.edu.vn (P. Pham), lt.kien@hutech.edu.vn (K.-T. Le), ngocluan192004@gmail.com (L. Bui), hq.bao@hutech.edu.vn (B. Huynh).

[1] https://www.ilo.org/resource/news/nearly-3-million-people-die-work-related-accidents-and-diseases.

[2] https://www.cpwr.com/research/data-center/the-construction-chart-book/chart-book-6th-edition-fatal-and-nonfatal-injuries-fatal-and-nonfatal-injuries-in-construction-and-other-industries/.

[3] https://www.bls.gov/news.release/osh.nr0.htm.

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0263-2241/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Beyond the United States, similar concerns have emerged from other industrialized regions. In the European Union, Eurostat recorded 3,286 fatal work-related accidents in 2022, with the construction sector accounting for a substantial proportion[4]. Ireland alone reported 11 construction-related deaths in 2023. In Asia, Japan recorded 223 workplace fatalities, and South Korea reported more than 2,000 occupational deaths, many of them linked to high-risk sectors. Singapore recorded 15 construction deaths in the second half of 2024, a surge that reignited national concern[5]. These reports collectively emphasize that construction safety challenges are not confined to any single region but demand international attention and innovation.

The continued prevalence of preventable accidents highlights a critical gap in the enforcement of, and compliance with, workplace safety measures, particularly those involving PPE. A 2020 report by the National Safety Council (NSC) found that nearly 36% of construction-site injuries could have been avoided through the use of appropriate PPE. Specifically, failure to use protective headgear contributed to 17% of traumatic brain injuries, while the absence of safety eyewear was implicated in more than 70% of eye-related incidents on-site[6]. Occupational Safety and Health Administration (OSHA) inspection data further reveal that PPE violations remain among the most frequently cited standards, with common infractions including failure to wear fall-protection harnesses, insufficient hand protection, and non-use of reflective vests in high-traffic areas[7]. These statistics reinforce the critical role of PPE in saving lives and highlight the urgent need for more effective compliance strategies, particularly within fast-paced, high-turnover job sites where human oversight is often inconsistent or insufficient.

Despite the recognized importance of PPE, conventional enforcement methods, such as manual inspections and safety briefings, remain constrained by their reliance on human vigilance, limited scalability, and inherent subjectivity. On large construction sites involving hundreds of workers, continuous PPE monitoring is difficult to sustain throughout the working day: unsafe behavior may occur between inspection rounds, and supervisors cannot always observe every worker across multiple work zones. Consequently, both researchers and industry practitioners have increasingly turned to artificial intelligence (AI) and computer vision (CV) as practical tools for automated safety monitoring. Compared with manual inspection, AI-based vision systems offer a more scalable, real-time alternative, particularly in environments where safety risks change rapidly and continuous supervision is otherwise difficult to maintain. Recent engineering studies have shown that vision-based intelligent systems can operate reliably under real industrial conditions. For instance, Mao et al. [2] successfully developed and deployed an AI-enabled visual defect detection system in ceramic manufacturing. The system demonstrated strong performance under varying illumination and surface conditions, challenges that closely resemble those encountered in construction-site surveillance.

Broader reviews by Yüksel et al. [3] and Puttegowda and Nagaraju [4] also show that modern engineering AI systems are moving beyond detection accuracy alone. Current research increasingly emphasizes real-time perception, interpretability, computational efficiency, and practical deployment in safety–critical or decision-sensitive environments. These directions are highly relevant to PPE monitoring, where a system must not only detect whether workers are wearing safety equipment but also issue timely warnings and support broader occupational health and safety management. Complementing these vision-based approaches, Zhang and Geng [5] proposed a directed, weighted complex-network-based methodology to identify high-risk causes of electrical personal accidents. By modeling accident causation chains and evaluating risk through frequency, severity, and occurrence probability, their study highlights the value of systematic, data-driven risk identification. Taken together, these works support the need for automated PPE monitoring frameworks that can detect compliance violations consistently, generate timely alerts, and be integrated into comprehensive safety management systems.

In recent years, considerable progress has been made in computer vision-based PPE detection. Several early approaches relied on classical object detection techniques, such as the histogram of oriented gradients (HOG) and Haar cascades, to recognize PPE items including helmets and safety vests s[6]. However, these techniques suffered from high false-positive rates and limited generalizability across diverse environments. The emergence of deep learning (DL) and its implementation [7,8] significantly advanced the field, with various models such as YOLO (You Only Look Once) [9], SSD (Single Shot MultiBox Detector) [10], and Faster R-CNN [11] gaining prominence for their superior detection accuracy and real-time inference capabilities. In addition, soft computing techniques, including fuzzy logic, evolutionary algorithms, hybrid methods, and other recent innovations, have also been applied to detection and tracking [12].

Dataset development has also played an important role in advancing PPE detection research. For example, the CHV dataset, introduced in 2021, featured five PPE classes: helmets, vests, masks, gloves, and safety glasses. This dataset was used to train a YOLOv5 [13] model that achieved an mAP of 86.55% on synthetic image sets. Building on this foundation, the CHVG dataset expanded both the dataset size and the diversity of object classes and employed the YOLOX architecture [14], pushing performance metrics to an mAP of 89.94%.

Moreover, recent studies have explored transformer-based detection frameworks, particularly DETR (DEtection TRansformer) [15], which showed promise in detecting PPE in crowded scenes and under occlusions, but at the cost of increased computational overhead and training complexity. For example, a comparative study found that while DETR-based models, such as the proposed DMS-DETR framework [16], outperformed YOLOv5 in small-object detection scenarios involving gloves and safety glasses, their inference time lagged by 37%, limiting their suitability for real-time applications on edge devices. Furthermore, the Fusion Detection Transformer (F-DETR) [17] introduced a heterogeneous multi-branch structure and represented an important step toward incorporating multi-scale features into an end-to-end DETR architecture.

At the same time, the YOLO family has continued to evolve and remains one of the most widely used choices for real-time PPE monitoring. Its popularity stems from its practical balance of detection accuracy, inference speed, and computational efficiency. As a result, YOLO-based models have been widely adopted in recent studies and prototype systems designed for dynamic and resource-constrained environments, including construction sites and industrial facilities [18–20]. Other efforts have focused on lightweight models optimized for embedded systems and Internet of Things (IoT) environments, such as lightweight convolutional neural network (CNN)–based models including MobileNet, Tiny-YOLO variants, YOLOv3, and YOLOv5, to enable low-latency detection in bandwidth-constrained settings [20–25].

Despite these technical advances, dataset quality remains one of the main obstacles to developing PPE detection systems that work reliably in real construction environments. Many commonly used datasets, including CHV, the Safety-Helmet-Wearing Dataset (SHWD), and HSEImages, are built from stock photographs, online images, or staged scenes. These datasets are useful for early model training, but they do not fully reflect the complexity of actual construction sites. Real surveillance footage is often far more complex: workers may appear at different distances from the camera, backgrounds can be crowded with

[4] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Accidents_at_work_statistics.

[5] https://www.mom.gov.sg/workplace-safety-and-health/wsh-reports-and-statistics.

[6] https://www.nsc.org/workplace/safety-topics/slips-trips-and-falls/slips-trips-and-falls-home?srsltid=AfmBOopLTJT_4SspXl6ck37fehbSu94qZwaFChYPFFxYDdWcCU2wFwHe.

[7] https://www.vectorsolutions.com/resources/blogs/top-10-osha-violations/.

equipment and materials, and lighting conditions may change across different areas of the site. Motion blur, occlusion, and low image quality are also common, yet these conditions are often limited or absent in public datasets.

A further issue concerns annotation detail. In many datasets, PPE labels remain too general: they may indicate whether a helmet or vest is present but often fail to capture more specific or practical compliance cases. For example, a worker may wear a helmet but no vest, wear safety glasses incorrectly, or have gloves that are only partly visible. Such distinctions matter in real safety monitoring, because compliance is not always a simple yes-or-no condition. Without more fine-grained annotations, models may learn to detect PPE objects yet still struggle to make reliable safety judgments.

Class imbalance further limits model performance. Large, visually obvious items, such as helmets and vests, usually appear more frequently than smaller items such as gloves and safety glasses. As a result, detection models may become biased toward the dominant classes and perform less consistently on underrepresented PPE categories. This is especially problematic because small PPE items are already harder to detect, owing to their limited pixel area and frequent occlusion. Another concern is generalization: a model trained on one dataset may perform well in that specific setting but may fail when applied to another construction site with different camera angles, lighting, worker uniforms, or background conditions. This lack of cross-domain validation suggests that current PPE detection systems still require more realistic, diverse, and site-independent evaluation before they can be considered reliable for practical deployment.

To address these challenges, recent studies have pursued three connected directions. The first is to expand PPE datasets to include small and underrepresented classes (e.g., gloves and safety glasses). The second is to improve multi-scale feature extraction for low-resolution surveillance imagery. The third is to introduce temporal or geometric reasoning to reduce unstable frame-level predictions. These directions are particularly important on real construction sites, where gloves, safety glasses, and safety boots may appear as tiny or partially occluded objects, and where a single false detection can trigger an incorrect compliance alert. However, most existing systems still treat PPE monitoring primarily as a static image-detection task, leaving lightweight deployment and stable PPE–person association across video frames insufficiently addressed. The present work therefore focuses on compact small PPE detection and temporal–spatial compliance validation under realistic CCTV conditions.

To overcome these limitations, this research proposes a comprehensive PPE detection framework built around a real-world dataset and a lightweight, small-object-oriented detector. The proposed dataset covers seven safety-related classes—safety boots, safety glasses, gloves, helmets, heads, persons, and reflective vests—captured from heterogeneous sources, including fixed CCTV cameras, mobile phones, drones, and public construction-scene images, under diverse lighting, camera viewpoints, and background conditions. Each image is annotated with bounding boxes and PPE category labels, supporting worker localization, PPE detection, and downstream compliance analysis. By including both worker-related classes, such as person and head, and PPE-related classes, such as helmet, glove, safety glasses, safety boots, and reflective vest, the dataset enables the proposed framework to infer PPE–-person relationships during post-processing, without relying on a heavy pose-estimation pipeline. Our main contributions in this paper are as follows:

- A geographically diverse Asia–Europe PPE dataset is constructed from real construction sites and public sources, capturing substantial variations in lighting conditions, worker poses, camera viewpoints, and scene complexity across seven safety–critical PPE categories.
- A fine-grained annotation framework is introduced to jointly represent worker-related regions and PPE categories, enabling PPE–person association and supporting compliance-oriented post-processing.
- A dedicated SmallPPE benchmark for extremely small PPE objects is developed using controlled data synthesis, photometrically consistent blending, and occlusion-aware augmentation, addressing a key limitation of existing PPE datasets in small-object representation.
- A lightweight small-object–oriented detection model, termed MicroPPE-Guard, is proposed, integrating an enhanced Feature Pyramid Network, Squeeze-and-Excitation–based feature fusion, and a decoupled detection head to improve detection accuracy under strict computational constraints.
- A Temporal–Spatial Compliance Module (TSCM) is introduced to improve measurement repeatability and temporal stability by smoothing confidence estimates, enforcing geometric consistency, and reducing spurious frame-to-frame fluctuations in video-based safety monitoring.
- An efficient framework balancing detection accuracy and computational overhead is established, demonstrating robustness and practical applicability in dynamic and uncertain industrial environments.

These contributions collectively show that MicroPPE-Guard addresses the study's main objective: reliable and real-time detection of extremely small PPE items, along with temporally stable PPE–person compliance reasoning under real-world construction-site surveillance conditions.

The remainder of this paper is organized as follows: Section 2 provides a comprehensive review of related work on CV-based PPE detection, including benchmark datasets, model comparisons, and deployment case studies. Section 3 details the research design, dataset construction methodology, and model training process. Section 4 presents experimental results, performance evaluations, and comparative analyses. Section 5 concludes the paper with a summary of contributions, implications for construction-site safety, and recommendations for future research.

2. Related works

Ensuring PPE compliance on construction sites has attracted sustained research interest, and a large body of work now applies computer vision and deep learning (DL) to automate worker safety monitoring. Because accident rates remain high and continuous manual inspection is impractical, automated PPE detection systems are increasingly regarded as practical tools for real-time safety supervision. This section reviews representative PPE detection datasets, small PPE detection methods, recent YOLO and Transformer-based construction-safety studies, and lightweight and temporal compliance-monitoring systems. Throughout, the discussion remains anchored to the central concerns of this study: accurate small PPE detection, efficient inference, and stable PPE–person association in real CCTV environments.

2.1. Deep learning-based PPE detection and benchmark datasets

Deep learning-based object detectors are widely used for PPE monitoring because they can localize workers and safety equipment in real time. Early studies concentrated on common, visually salient classes such as helmets, vests, and persons. Z. Wang et al. introduced the CHV dataset and trained a YOLOv5-based detector [13], achieving an mAP of 86.55% for helmet, vest, and person detection. Although this dataset provided a valuable benchmark for construction-site safety monitoring, its narrow class coverage and limited environmental variation make it less representative of the complex conditions encountered in real-world surveillance.

To address this, the CHVG dataset [14] extended the class set to include safety glasses and uncovered heads, achieving 89.94% mAP with YOLOX. Even so, both datasets remained centered on common PPE items and omitted less frequent but safety–critical equipment such as

gloves and ear protection. Ahmad and Rahimi pushed class diversity further with the SH17 dataset [26], which comprises 17 PPE-related categories. Yet even broader class definitions do not resolve two persistent limitations: small PPE items such as gloves, safety glasses, and boots remain underrepresented, and occlusion, low resolution, illumination variation, and cluttered construction backgrounds continue to degrade detection reliability.

Overall, existing PPE datasets have advanced the field but remain biased toward larger, more visible safety items. This bias opens a gap between benchmark performance and practical CCTV deployment, where PPE objects may span only a few pixels and workers are frequently partially occluded or viewed from elevated angles. Such limitations motivate datasets that deliberately include small PPE categories, diverse construction environments, and evaluation settings that more closely reflect real surveillance conditions.

2.2. Small PPE detection in real CCTV systems

In practical construction and industrial surveillance, detecting small PPE items—especially gloves, safety glasses, and safety boots—remains a significant technical challenge. Such items are typically captured at low resolution because CCTV cameras are mounted at a distance or overhead, which yields limited pixel coverage, frequent occlusion, and reduced detectability under low-light or visually cluttered conditions. Compared with larger equipment such as helmets or reflective vests, small PPE is therefore considerably harder to detect reliably.

To address these challenges, recent studies have proposed both architectural and data-centric innovations. An improved YOLOX-based framework [27] developed for tunnel construction environments introduced ConvNeXt modules, an additional detection head, and CLAHE-based enhancement to strengthen multi-scale feature extraction and low-light visibility. This design achieved a 7.17% improvement in average precision for small PPE classes over the standard YOLOX baseline while sustaining real-time performance at 22 FPS, and it was accompanied by a dedicated low-light dataset containing over 6,800 small-object instances. In a complementary direction, Al-Azani et al. [49] developed a large-scale end-to-end PPE compliance monitoring system that leveraged heterogeneous surveillance data to benchmark multiple detection architectures, showing that YOLO-based detectors offer superior accuracy and generalization under real-world conditions when deployed in GPU-accelerated real-time pipelines.

Beyond PPE-specific research, industrial inspection studies offer useful insight into how small objects can be detected under visually constrained conditions. This is directly relevant to PPE monitoring, where items such as gloves, safety glasses, and masks often appear small, partially occluded, or blurred in surveillance footage. Zhang et al. [28] proposed a hybrid detection–segmentation framework that couples an enhanced YOLOv8 detector with attention-guided semantic segmentation, enabling fine-grained analysis of the position and size of visually subtle defects. This suggests that region-level segmentation can be advantageous when objects occupy only a few pixels and standard bounding-box detection may not provide enough spatial detail. Addressing scale sensitivity more directly, Han et al. [29] introduced a Scale-Aware Feature Pyramid Network (SA-FPN) that reduces semantic inconsistency across pyramid levels through attention-guided feature fusion, markedly improving small-object localization across multiple benchmarks. Complementing these accuracy-focused designs, Bin et al. [30] proposed CI-YOLO, a lightweight YOLOv8-based architecture optimized for long-distance, low-resolution inspection that helps improve recall and mAP while substantially reducing computational complexity.

Taken together, these studies indicate that robust small-object detection rests on three factors: preserving high-resolution spatial cues, strengthening multi-scale feature fusion, and maintaining a lightweight inference pipeline. Most existing solutions, however, remain either task-specific or only partially aligned with PPE compliance monitoring. Industrial inspection methods are not designed to associate PPE with individual workers, and many PPE-specific detectors still emphasize helmets and vests over tiny items such as gloves, safety glasses, and boots. Reliable small PPE monitoring in construction scenes therefore requires not only stronger feature extraction but also a compact architecture suited to real-time deployment.

2.3. Recent YOLO and transformer-based construction-safety detection

Recent construction-safety studies have increasingly turned to YOLOv10 and Transformer-enhanced detectors, reflecting a broader shift toward stronger feature representation under realistic camera conditions. S. Wang [31] evaluated YOLOv10 variants with Transformer backbones for automated non-PPE detection across surveillance images, body-worn camera footage, and public benchmark datasets, covering the non-helmet, non-mask, non-vest, non-glove, and non-shoes categories. The Swin Transformer-based YOLOv10 gave the strongest overall result, with AP_{50} values of 92.40% for non-helmet, 88.17% for non-mask, 87.17% for non-vest, 85.36% for non-glove, and 83.48% for non-shoes, yielding an average AP_{50} of 87.32%; the study further reported an $AP_{50:95}$ of 64.71% and an inference speed of 35.32 FPS. These results indicate that Transformer-enhanced YOLOv10 can balance accuracy and real-time performance for non-PPE monitoring. Notably, the lower scores on smaller categories such as non-glove and non-shoes confirm that fine PPE-related violations remain harder to detect than larger visual targets.

Safety helmet monitoring has likewise been revisited with YOLOv10 and advanced Transformer backbones. S. Wang et al. [32] studied helmet and non-helmet detection under both surveillance and body-worn camera settings, reporting that Transformer-based YOLO models improve robustness across differing camera viewpoints. This is a relevant direction, since construction monitoring rarely depends on a single image source. Nonetheless, the task remains largely helmet-centered and does not fully cover multi-class small PPE compliance involving gloves, safety glasses, boots, vests, and worker-level association.

Although not concerned with PPE directly, Eum et al. [33] applied YOLOv10 with Transformer architectures to heavy-equipment detection on construction sites. Their work illustrates the growing adoption of Transformer-enhanced YOLO designs in construction-safety perception, particularly for complex scenes with wide variation in target scale and background clutter. Heavy-equipment detection nonetheless differs from worker-level PPE monitoring: the targets are generally larger, are less attached to human body regions, and require no PPE–person compliance association.

Cross-domain reliability has been tackled more directly by S. Wang [34], who proposed a domain-adaptive Faster R-CNN framework for non-PPE identification from body-worn and general images. The method combined image-level and instance-level adversarial domain classifiers, linked through gradient reversal layers, to learn domain-invariant features for five non-PPE categories: non-helmet, non-mask, non-glove, non-vest, and non-shoes. With a ResNet-152 backbone, augmented source and target data, and tuned hyperparameters, the best model reached 86.84% mAP on unseen construction-site images and improved performance by up to 14 mAP points over supervised Faster R-CNN baselines. These results confirm that domain shift among general imagery, body-worn cameras, and construction-site images is a serious obstacle for non-PPE detection. The framework nevertheless remains a two-stage detector and is not primarily designed for compact real-time CCTV deployment or temporal compliance reasoning.

Collectively, these studies reflect the prevailing trend in construction-site safety monitoring: stronger YOLO versions, Transformer-enhanced backbones, and domain-adaptive training are being enlisted to improve non-PPE recognition. Their emphasis, however, differs from that of the present work. Transformer-based methods can raise detection accuracy but often at the cost of increased model complexity, whereas domain-adaptive Faster R-CNN prioritizes cross-

source generalization over lightweight deployment. More importantly, most recent non-PPE studies center on violation categories, helmet-focused monitoring, heavy equipment, or camera-domain transfer, leaving fine-grained small PPE items—especially gloves, safety glasses, and boots—less comprehensively addressed under low-resolution CCTV constraints.

2.4. Lightweight and temporal PPE compliance monitoring

Research on automated PPE compliance monitoring has advanced rapidly toward systems that integrate pose estimation, temporal modeling, small-object detection, super-resolution enhancement, and lightweight edge deployment. These developments reflect a growing recognition that reliable monitoring in complex industrial environments requires more than static bounding-box detection; it demands a holistic understanding of human posture, temporal consistency, and robustness under diverse real-world conditions.

A major breakthrough in integrated compliance systems is the coupling of pose estimation with PPE detection. López et al. [35] introduced a comprehensive pipeline that fuses InternImage-L worker detection, YOLOv7 PPE detection, ViTPose human pose estimation, and pose-guided duplicate elimination, enabling robust compliance verification even under occlusion and in dense worker crowds. Their system achieved strong recall and F1 scores, but it still depends on medium-range, clear-visibility cameras, which limits its effectiveness for the long-distance CCTV commonly used on large construction sites.

Temporal reasoning has also gained prominence. Zhang et al. [36] developed a video-based helmet-use monitoring framework that combines YOLOv9, YoloPose, and StrongSORT tracking with helmet–head region matching and temporal state tracking. Their method achieved 98.89% accuracy, illustrating the value of temporal smoothing when workers move rapidly or are briefly obscured. As with many temporal systems, however, performance degrades under extreme viewpoint changes or at very small object sizes.

In a closely related tracking-based PPE monitoring study, Mahadi et al. [37] implemented a YOLOv8 and DeepSORT framework for personal safety equipment detection and tracking, using YOLOv8 for real-time PPE recognition and DeepSORT for object-level association across frames on a custom dataset. This work demonstrates the practical feasibility of pairing modern one-stage detectors with tracking-by-detection methods for occupational safety monitoring. Its emphasis, however, is on general PPE tracking—particularly visually salient items such as helmets and vests—rather than the fine-grained localization of very small PPE categories such as gloves, safety glasses, and safety boots under long-distance CCTV conditions. MicroPPE-Guard, by contrast, targets small PPE-oriented data construction, enhanced multi-scale feature fusion, and lightweight temporal–spatial compliance reasoning without relying on a heavier full tracking pipeline.

In parallel with pose and temporal modeling, researchers have explored zero-shot and vision–language-based PPE compliance to reduce reliance on large labeled datasets. Gil and Lee [38] proposed a zero-shot framework that detects workers, crops body-part regions, generates captions, and matches them against PPE-related prompts. Without any fine-tuning, the system reaches an 82.6% F1 score on hardhat monitoring, indicating strong potential for rapidly deployable systems. Even so, zero-shot methods still struggle with fine-grained PPE, particularly when items occupy only a few pixels.

Small-object detection has emerged as one of the most critical challenges in PPE compliance research. Li et al. [39] introduced OAM-YOLO, which incorporates attention-enhanced multi-scale fusion to target PPE items such as gloves, safety glasses, and safety boots. Their model improves average precision for small PPE while maintaining real-time throughput, directly addressing the limitations of general-purpose detectors. Complementing this line of work, the ESPCN-YOLO framework [40] integrates super-resolution (ESPCN) with YOLOv8, markedly enhancing PPE detection in low-light, long-distance, and extremely small-object scenarios—conditions that closely mirror real construction CCTV. In a similar vein, SD-YOLOv5s [41] incorporates DilateFormer attention and a dedicated small-object detection branch, achieving approximately 93.7% AP on complex construction scenes and further confirming that specialized architectures are needed for fine-grained PPE categories.

Beyond accuracy, recent studies increasingly emphasize real-time capability and deployability on edge devices. Kim et al. [42] demonstrated an improved YOLOv8-based system for fall-from-height prevention that runs on edge hardware with low latency. Likewise, a 2025 implementation of YOLOv5-Lite [43] targets lightweight helmet detection for live construction-site monitoring, showing that optimized lightweight models can attain practical inference speeds without substantial accuracy loss. Together, these works underscore the industry's demand for solutions that are not only accurate but also computationally efficient for field deployment.

The broader context of these developments has been synthesized in a systematic review by Vukicevic et al. [44], which identifies persistent weaknesses across state-of-the-art PPE compliance systems: limited geographic diversity in datasets, poor small-object performance, sensitivity to illumination variation, domain shift across sites, and limited temporal robustness. These observations recur across the specialized works reviewed above and underscore the field's continued reliance on near-ideal imaging conditions.

Collectively, the recent literature demonstrates substantial progress toward integrating pose estimation, temporal stability, small-object modeling, and edge-friendly inference. Yet a recurring limitation persists: lightweight and temporally aware systems still struggle to sustain accuracy on small PPE items under long-distance, low-resolution CCTV. These persistent shortcomings directly motivate our development of a geographically diverse Asia–Europe PPE dataset and the MicroPPE-Guard framework, which together aim to deliver high small-object accuracy, temporal reliability, and edge-device feasibility for real-world construction environments.

3. Research gap and position of MicroPPE-Guard

The reviewed literature shows that PPE monitoring has gradually moved beyond simple helmet- and vest-detection tasks toward broader directions, including multi-class PPE recognition, small-object enhancement, Transformer-based detection, domain adaptation, pose-assisted reasoning, and temporal tracking. Important gaps nonetheless remain. Small PPE categories—especially gloves, safety glasses, and safety boots—are often poorly represented in public datasets. Moreover, many high-performing Transformer-based or pose-assisted models are difficult to deploy on ordinary CCTV infrastructure, as their computational cost is frequently too high for low-power or edge-based monitoring. A further limitation is methodological: most studies still evaluate PPE detection at the static frame level, whereas real compliance monitoring requires the system to associate each detected PPE item with the correct worker consistently over time. Viewpoint changes, poor lighting, video compression, and site-specific background variation further erode model reliability when systems trained on curated datasets are transferred to real construction environments.

MicroPPE-Guard is developed in response to these limitations. Rather than optimizing static detection accuracy alone, the proposed framework jointly addresses small PPE localization, lightweight computation, and temporally stable compliance reasoning. The SmallPPE-oriented data construction improves learning for tiny, visually weak safety items; the SE-enhanced feature pyramid helps preserve scale-sensitive cues that might otherwise vanish in low-resolution CCTV footage; and the decoupled detection head sharpens the balance between category classification and bounding-box localization. Beyond detection, the Temporal–Spatial Compliance Module suppresses unstable frame-level outputs and removes geometrically implausible PPE–person assignments. The novelty of MicroPPE-Guard therefore does not

rest on any single component viewed in isolation; its principal value lies in how these components are adapted and integrated into a compact small PPE compliance framework designed specifically for low-resolution, cluttered, edge-deployed CCTV construction environments.

4. Proposed method

This section describes the methods and procedures adopted in our study, covering dataset collection and preprocessing, annotation, model selection, and the experimental setup.

4.1. Object identification

The primary aim of this research is to build a dataset tailored to real-world construction-site environments, in which the selected object categories both represent commonly used PPE and carry high practical relevance. Each category must also occur frequently enough in the collected data to permit substantial sampling, which in turn enables comprehensive model training.

We identified seven key object classes:

- Person: Denotes a construction worker and provides the primary human region for associating PPE items with individual workers.
- Head: Denotes the worker's head region and supports helmet-use verification by enabling the model to distinguish exposed heads from helmet-covered heads.
- Helmet: Denotes protective headgear worn to reduce the risk of head injuries on construction sites. In the annotation files, this class is encoded as hat.
- Glove: Denotes hand protection worn to reduce injuries from cuts, impacts, heat, chemicals, and other hazardous materials.
- Safety glasses: Denotes eye protection worn to shield workers from dust, chemical splashes, sparks, and flying debris. In the annotation files, this class is encoded as glasses.
- Safety boots: Denotes protective footwear designed to reduce injuries from falling objects, sharp materials, slippery surfaces, and electrical hazards. In the annotation files, this class is encoded as boot.
- Reflective vest: Denotes high-visibility clothing worn to improve worker visibility in construction zones, traffic-adjacent areas, and low-light working environments. In the annotation files, this class is encoded as vest.

The choice of these categories reflects both practical considerations and the feasibility of collecting sufficient data for accurate labeling of each object type. Less relevant PPE categories, which are uncommon on construction sites, were excluded so as to concentrate detection on the equipment most essential to worker safety.

4.2. Dataset collection and preparation process

To build a dataset that faithfully reflects the visual complexity of real construction environments, we designed a multi-stage data-collection pipeline that combines field-recorded imagery, diversified public sources, and synthetically generated SmallPPE data. Fig. 1 provides an overview of this pipeline, spanning acquisition, filtering, preprocessing, and annotation. This design grounds the dataset in authentic on-site conditions while broadening its visual diversity through external material. In particular, although most field recordings were captured at Asian locations, incorporating publicly available construction images from both Asia and Europe widened the range of backgrounds, worker appearances, and scene compositions represented in the dataset, enabling the model to generalize more effectively across geographic regions.

To ensure that both the field-recorded and crawled images met consistent standards of quality and relevance, we defined a concise set of selection criteria to guide acquisition and curation. These criteria were designed to secure geographic diversity across Asia and Europe, capture a wide range of lighting conditions, and represent natural worker activities, including challenging cases such as small-scale PPE and partial occlusions. They also shaped the keyword strategy used during web crawling to retrieve relevant external images. Table 1 summarizes the main selection criteria, their underlying rationale, and the keyword filters applied during dataset construction.

Self-collected data were acquired with handheld mobile devices during actual construction activities, capturing workers in natural, unscripted movements such as walking, lifting, turning, bending, and handling tools or materials. These recordings span a broad range of camera viewpoints and distances, yielding both close-up and mid-range samples. Beyond still images, we recorded 10 short video sequences (≈5 min each) documenting workers' continuous motion, thereby enriching the dataset with pose transitions and movement patterns that are difficult to obtain from still images alone.

To enhance transparency and reproducibility, the final proposed 25K dataset was organized by three major data sources, as summarized in Table 2. It comprises self-collected construction imagery, public Asia–Europe construction images, and synthetic SmallPPE samples

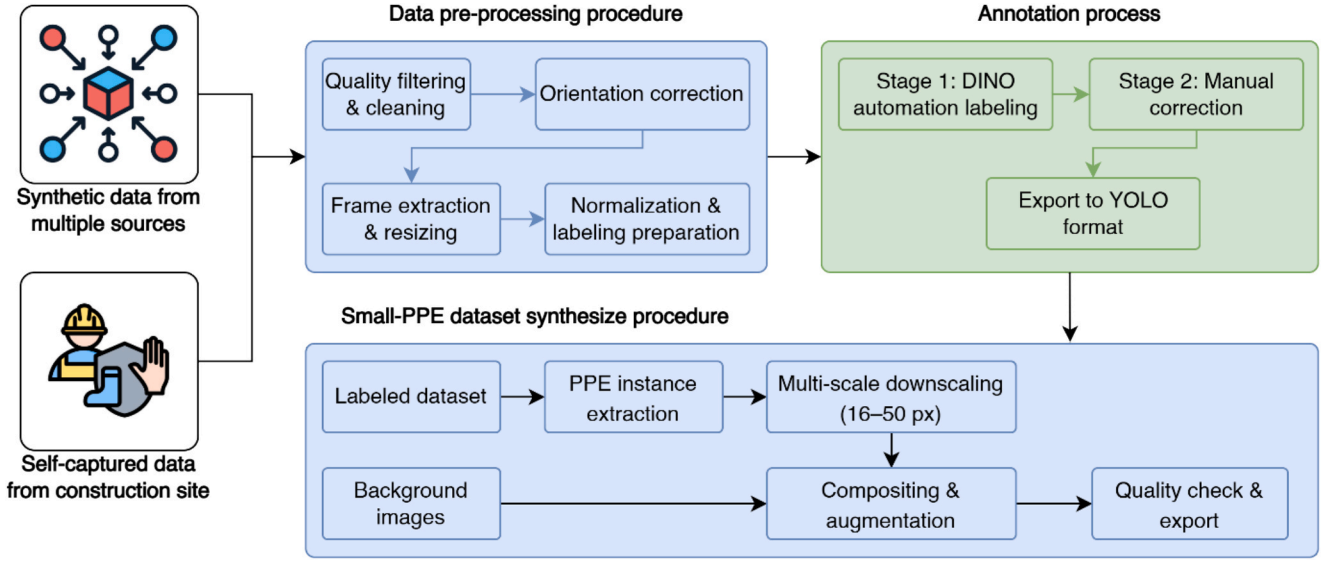


Fig. 1. The integrated pipeline, combining synthetic data, self-captured construction imagery, preprocessing, and two-stage labeling.

Table 1
Criteria for selecting and curating images from asia–europe construction sources.

Criterion	Purpose / rationale	Keywords / filters for crawling
Geographic Diversity (Asia + Europe)	Improve cross-regional generalization; include different PPE colors, clothing styles, and construction layouts.	"Asia construction site," "Europe PPE workers," "工地 安全帽," "chantier PPE Européen"
Environmental & Lighting Variety	Ensure robustness under sunlight, glare, shadows, dust, cloudy weather, and indoor/outdoor conditions.	"low light construction," "dusty site PPE," "night shift workers," "backlit construction"
Worker Pose & Activity Diversity	Capture realistic movements (walking, bending, lifting, tool handling) for better model behavior understanding.	"construction worker lifting," "bending worker," "operating tools PPE"
Viewpoint & Camera Diversity (CCTV + handheld)	Model should handle long-range CCTV small objects and near-range mobile captures.	"CCTV construction safety," "surveillance PPE," "wide-angle construction view"
PPE Visibility & Occlusion Cases	Ensure gloves, safety glasses, and safety boots appear in both clear and partly occluded conditions; include very small PPE (<20 px).	"small object PPE," "occluded PPE," "gloves far distance," "tiny PPE detection"

Table 2
Composition of the proposed 25 K dataset by data source.

Data source	Percentage	Approximate number of images	Description
Self-collected construction images and video-derived frames	30%	7,500	Captured from real construction activities using handheld devices and short video sequences, covering natural worker poses, motion patterns, camera viewpoints, and site-specific conditions.
Public Asia–Europe construction images	50%	12,500	Collected from diversified public construction sources to increase geographic, environmental, viewpoint, lighting, and scene diversity.
Synthetic SmallPPE images	20%	5,000	Generated from manually verified PPE crops and composited into realistic construction backgrounds to enrich tiny, distant, and partially occluded PPE cases.
Total	100%	25,000	Final curated dataset used for PPE detection, small-object evaluation, and robustness assessment.

generated to strengthen the representation of tiny and partially occluded PPE objects.

The dataset was partitioned into training, validation, and test subsets using a 70/20/10 split. To prevent data leakage, the partition was performed at the source level rather than by random image sampling. For video-derived data, all frames extracted from a given source video were assigned to a single subset, so that visually similar adjacent frames did not span the training, validation, and test sets. Likewise, public images drawn from the same source page or visually similar scene group were kept within one subset, and synthetic SmallPPE samples derived from the same original PPE crop were placed in the same subset as their source instance. This strategy reduces near-duplicate leakage and yields a more reliable estimate of model generalization.

After collection, all images and videos were carefully reviewed so that only usable and relevant samples were retained. Images that were blurred, duplicated, poorly framed, or unrelated to construction-site safety were removed during filtering, yielding a cleaner and more reliable dataset for training. Representative samples from the curated dataset are shown in Fig. 2, illustrating the diversity of construction-site scenarios, worker activities, camera viewpoints, and environmental conditions covered in this study. The data predominantly depict workers wearing complete PPE in realistic construction environments, allowing the dataset to closely reflect practical monitoring conditions. They also span a wide range of worker postures and activities—standing, walking, bending, sitting, handling tools, and working collaboratively—captured at varying camera distances and viewing angles.

Once curated, the dataset underwent preprocessing to standardize the samples and prepare them for training. First, auto-orientation correction was applied so that all images were properly aligned; this step prevents errors arising from unintended rotation or flipping and helps the model learn PPE features more consistently across scenes. Next, all images were resized to 1024 × 1024 pixels to provide a uniform input resolution, which improves training stability, simplifies batch processing, and reduces unnecessary computational variation. For video data, frames were extracted at one frame per second—a rate chosen to capture meaningful changes in worker movement and interaction while avoiding excessive redundancy between consecutive frames. The resulting dataset therefore offers both visual diversity and sufficient temporal coverage, both of which are important for developing PPE detection models intended for real-time construction-site monitoring.

Following preprocessing, the dataset was annotated using a two-stage labeling pipeline designed to maximize both accuracy and efficiency.

5. Stage 1: Automatic labeling using DINO model

In the first stage, initial bounding boxes and class labels were generated with Grounding DINO [45], a transformer-based open-set detection model, and were then subjected to complete manual verification and correction. Grounding DINO localizes and labels objects from text prompts, providing reliable initial annotations for subsequent refinement.

6. Stage 2: Manual review and adjustment

Following automatic labeling, all images and video frames underwent meticulous manual review and refinement. This step ensured that the labels were accurate and consistent with safety standards and research requirements. Each annotated instance was carefully inspected to verify its integrity, and any errors introduced during automatic labeling were corrected; missing labels were also added by hand to maximize dataset precision. After this process, the dataset comprised a total of 25,000 images. Further details are provided in Fig. 3.

In addition to the main dataset, a dedicated synthetic SmallPPE subset was constructed to address one of the most challenging cases in real CCTV-based monitoring: PPE items that appear extremely small in the image. Once the main annotation was complete, high-quality PPE instances were selected from the manually verified images—namely safety boots, safety glasses, gloves, helmets, reflective vests, and persons. These objects were then cropped and resized to emulate the appearance of PPE captured by long-distance surveillance cameras.

To simulate this condition, each PPE crop was downscaled by fixed ratios, such as × 0.25, ×0.125, and × 0.1. The resized instances produced target boxes with longer-edge sizes of approximately 16–50 pixels, representing tiny-object conditions commonly encountered in

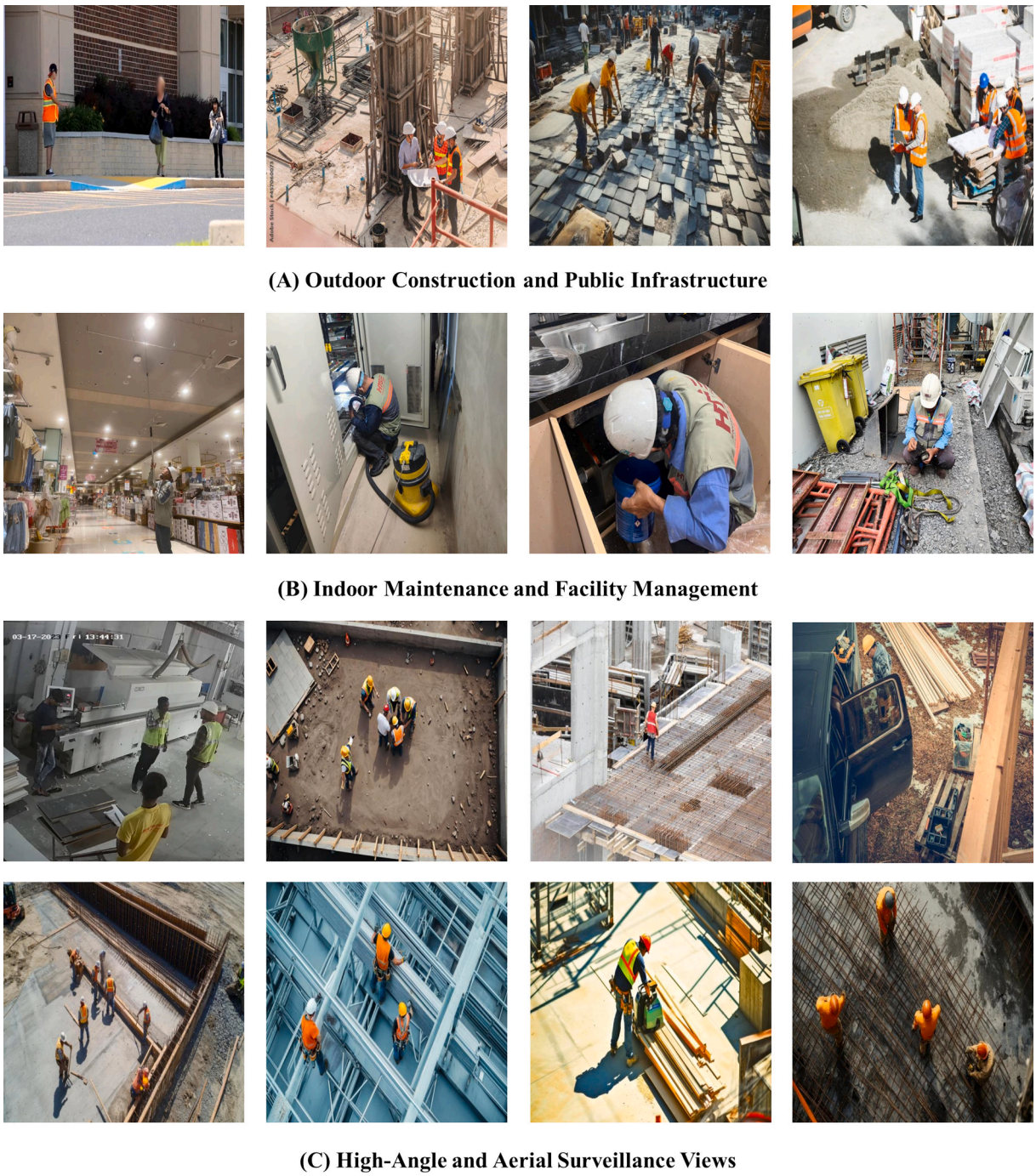


Fig. 2. Representative construction-site images collected during data acquisition, illustrating diverse worker poses, activities, and environmental conditions across a range of viewpoints and task contexts.

low-resolution CCTV imagery. The resized objects were then inserted into background images drawn from two sources: field-recorded construction-site scenes and publicly available Asian and European construction photographs selected according to the criteria in Table 1. Combining these sources lets the subset span a wider range of environments, including outdoor sites, indoor construction spaces, and scenes with poor or adverse visibility.

To make the synthetic images as realistic as possible, each was produced through a multi-step blending workflow. PPE patches were first placed at random, subject to simple spatial rules that avoid implausible results (e.g., attaching items to plausible body regions or textured surfaces). Photometric transformations—brightness jitter, gamma correction, color-temperature shifts, and shadow simulation—were then applied to match the background lighting. These were followed by geometric distortions, including slight rotation (±8°), perspective warping, partial occlusion (random rectangles or tool-like shapes), and micro-motion blur (1–3 px), to reproduce CCTV compression or subject movement. Finally, alpha-guided Poisson blending and noise injection (JPEG compression noise, Gaussian noise) were applied so that the inserted PPE patches matched the texture and overall visual statistics of the background.

From each original PPE instance, 8–12 synthetic samples were generated using different scales, backgrounds, and augmentation settings. The resulting subset provides a concentrated distribution of hard-

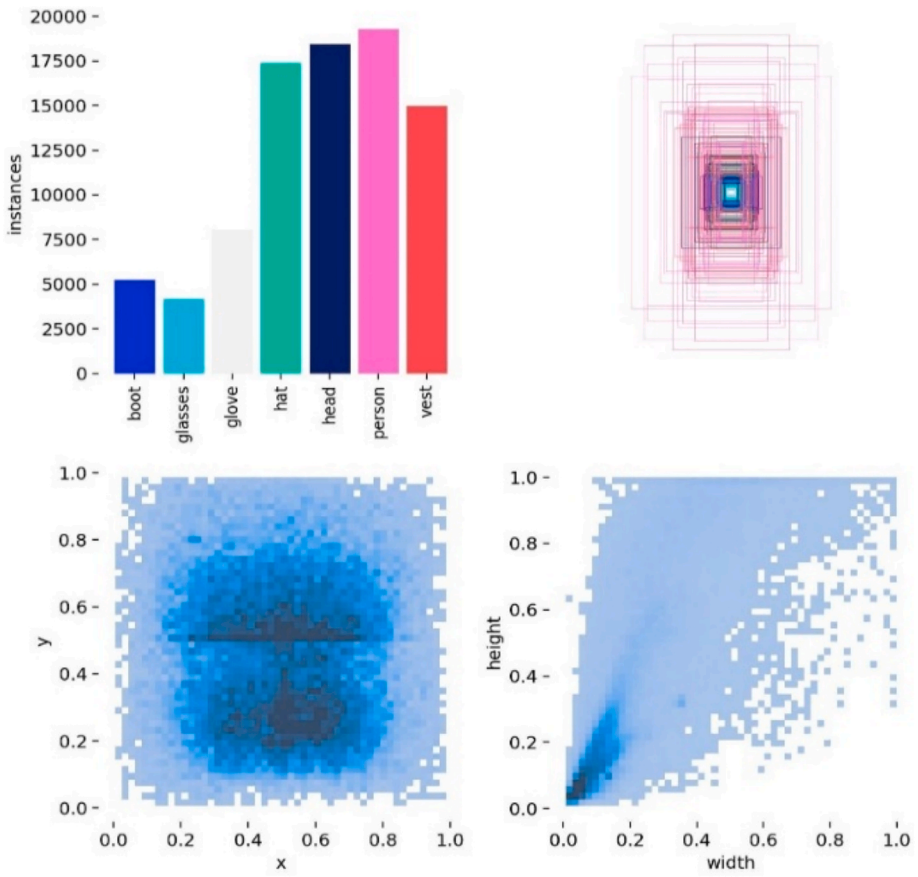


Fig. 3. Visualization of PPE dataset annotations, showing class instance counts (top left), normalized bounding-box overlays (top right), the spatial distribution of object centers (bottom left), and the bounding-box size distribution (bottom right).

to-detect PPE examples, substantially enriching the long-tail portion of the dataset that natural images rarely capture. It thus constitutes a challenging benchmark, curated specifically to evaluate small-object performance and to improve the robustness of models deployed in real-world CCTV-based safety monitoring systems.

6.1. Proposed MicroPPE-Guard

The core of our proposed method is MicroPPE-Guard, a custom adaptation of the YOLO detection framework tailored to the challenges of PPE monitoring on real-world construction sites. At a high level, MicroPPE-Guard ingests an input image $I \in R^{H\times W\times 3}$ and produces a set of N detections $\{(b_i, c_i, s_i)\}$, where each bounding box $b_i = [x_i, y_i, w_i, h_i]$ encodes normalized center coordinates and dimensions, $c_i \in 1, \cdots, 7$ corresponds to one of our seven object classes, and $s_i \in [0, 1]$ is a confidence score. To accomplish this with high precision, especially for small objects such as gloves and safety glasses, MicroPPE-Guard integrates four tightly coupled modules (See Fig. 4):

1. Multi-scale backbone with an enhanced Feature Pyramid Network (FPN) [46].
2. Channel-wise attention fusion block based on Squeeze-and-Excitation recalibration [47].
3. Decoupled dual YOLO head for classification, objectness, and regression.
4. Temporal–Spatial Compliance Module (TSCM) for post-processing stability and geometric validation.

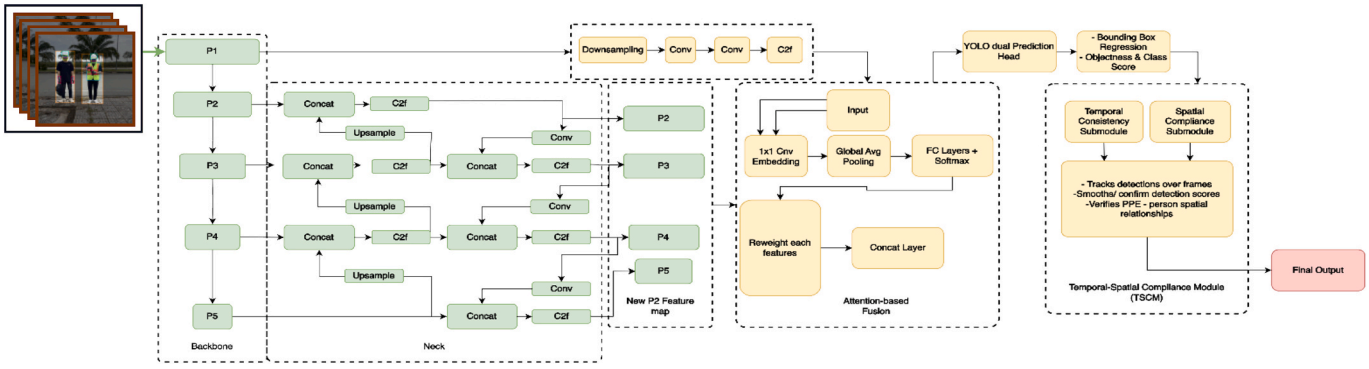


Fig. 4. Proposed MicroPPE-Guard architecture.

As summarized in Table 3, existing PPE detection systems typically optimize either detection accuracy or deployment efficiency, but rarely both under realistic CCTV constraints. YOLO-based variants such as MARA-YOLO and MKD-YOLO [18,21] improve multi-class detection and model compactness, yet they are mainly evaluated on medium-scale, well-lit scenes where helmets and vests dominate the annotations, with limited emphasis on tiny PPE such as gloves or safety glasses. Large-scale systems and pose-assisted frameworks [35] provide more complete compliance reasoning, but they rely on high-resolution imagery and heavy multi-branch architectures that are difficult to deploy on edge devices. Recent video-based transformer models further enhance temporal consistency but at the cost of substantial computational overhead. In contrast, MicroPPE-Guard is explicitly designed around three requirements: (i) robust detection of extremely small PPE using an enhanced FPN and SE-based fusion tailored to our SmallPPE subset; (ii) temporal and geometric compliance reasoning via a lightweight Temporal–Spatial Compliance Module (TSCM); and (iii) practical deployability on resource-constrained CCTV hardware, achieved through a compact YOLO-style backbone and minimal post-processing. The novelty lies in integrating these elements into a unified small PPE compliance framework: the SmallPPE synthesis procedure—a data-construction and training component, not an architectural module—strengthens tiny-object learning, while the detector architecture, comprising the enhanced FPN, Squeeze-and-Excitation–based feature fusion, a decoupled detection head, and the TSCM, preserves scale-sensitive features and suppresses unstable or geometrically implausible frame-level predictions. The inclusion of TSCM is motivated by evidence from both PPE-specific and generic video-detection literature. video-based PPE systems such as Yang et al. [48] already show that explicitly modeling temporal context reduces frame-to-frame prediction noise and improves compliance estimation, although their transformer backbone is computationally expensive. More generally, prior object-detection studies indicate that enhancing contextual feature representation through multi-scale aggregation and improved feature fusion can improve detection robustness and reduce unstable predictions in challenging surveillance imagery [40,41]. Inspired by these findings, TSCM adopts a minimalist variant of temporal aggregation: it combines exponential moving average smoothing of per-track confidence scores with IoU-based Hungarian matching and a geometric PPE–person overlap constraint. This design brings many of the benefits of temporal modeling, such as stability, reduced spurious alerts, and improved compliance reasoning, while keeping the computational footprint small enough for real-time execution on edge-deployed CCTV systems.

First, given an input image $I \in \mathbb{R}^{H\times W\times 3}$, we extract hierarchical features using a CSPDarknet-style [50] backbone pretrained on ImageNet [51]. This produces feature maps:

$$C_\ell \in \mathbb{R}^{\frac{H}{2^\ell}\times\frac{W}{2^\ell}\times D_\ell} \quad (\ell = 1,\cdots,5) \tag{1}$$

where early layers (ℓ=1, 2) retain fine spatial detail for small-object localization, and deeper layers (ℓ=4, 5) capture the semantic context necessary to distinguish PPE from background structures. To fuse these disparate scales, we employ an enhanced FPN: each C_ℓ is first projected via a 1×1 convolution into a uniform channel dimension D, yielding lateral features L_ℓ. We then build top-down feature maps P_ℓ recursively:

$$P_5 = L_5, P_\ell = \mathrm{Conv}_{3\times3}([\mathrm{Upsample}(P_{\ell+1}), L_\ell])(\ell = 4,3,2) \tag{2}$$

where nearest-neighbor upsampling doubles the spatial dimensions and the 3×3 convolution merges semantic and localization cues. This top-down, lateral-connection strategy preserves the high-resolution detail essential for detecting objects as small as a glove fingertip (16×16 pixels) while leveraging deeper context to reduce false positives in cluttered construction backgrounds.

The enhanced FPN generates multi-scale feature maps through top-down and lateral fusion, which are then resized to a common resolution and concatenated along the channel dimension within the SE fusion block. To allow the network to learn which scales are most informative for each sample and PPE class, we integrate a Squeeze-and-Excitation (SE) attention module [52]. We first upsample all P_ℓ to a common resolution $\frac{H}{4}\times\frac{W}{4}$ and concatenate them along the channel axis:

$$X = \left[\widetilde{P}_2, \widetilde{P}_3, \widetilde{P}_4, \widetilde{P}_5\right] \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times 4D} \tag{3}$$

Global average pooling reduces X to a channel descriptor $z \in \mathbb{R}^{4D}$, which passes through a two-layer bottleneck MLP with reduction ratio $r = 16$:

$$a = \sigma(W_2\mathrm{ReLU}(W_1 z)) \in (0,1)^{4D} \tag{4}$$

Because it applies channel-wise rather than spatial attention, this fusion block recalibrates the relative importance of the multi-scale feature channels without directly suppressing small PPE locations in the spatial map.

This attention vector reweights each channel of X by simple multiplication $X'_c = a_c X_c$. A final 1×1 convolution projects X' into the fused feature map:

$$F \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times D_f} \tag{5}$$

where $D_f = 256$. By adaptively amplifying the scales most relevant to helmet recognition or glove detection in each scene, SE fusion is

Table 3
Comparative analysis of MicroPPE-Guard against representative PPE-detection frameworks.

Study / framework	Backbone & architecture	Small-object strategy	Temporal reasoning	Dataset scale & diversity	Deployment suitability	Limitations vs. Ours
López et al. (2025) [35]	InternImage-L + ViTPose + YOLOv7	None (focus on larger PPE, posture cues only)	Basic pose-based heuristics	Industrial indoor scenes; limited variation	Medium-weight GPU	Weak tiny-PPE accuracy; indoor bias; no CCTV long-distance handling
Yang et al. (2023) [48]	Spatiotemporal Transformer	Does not specifically address PPE	Full transformer temporal attention	Surveillance video only; lacks cross-region diversity	Very heavy; not edge-suitable	High compute cost; tiny PPE still poorly detected
Zhang et al. (2025) [36]	YOLOv9 + YoloPose + StrongSORT	Pose-guided spatial association	Temporal state tracking across frames	Helmet-focused dataset; limited scene diversity	GPU needed	Sensitive to missing keypoints and heavy occlusion; limited to helmet only or tiny-object specialization
Z. Wang et al. (2023) [27]	Improved YOLOX + ConvNeXt	Small-object anchors	None	Low-light tunnel dataset	Moderate	Tunnel-specific; low generalization to open outdoor sites
Al-Azani et al. (2024) [49]	CNN/YOLO Variants	Implicit small-object handling via scale diversity	None	Very large CCTV dataset; fixed-view, distance-dominated	GPU servers	Poor performance for objects < 20 px; heavy inference
MicroPPE-Guard (Ours)	YOLO-based + SE-FPN + Dual Head	Small-PPE Synthesis (16–50 px) + SE attention	TSCM: EMA smoothing + Hungarian matching	Asia–Europe mixed, real-site, multi-lighting, varied behavior	Lightweight, potentially for edge deployment	Addresses several key gaps related to small-PPE

intended to improve detection robustness under occlusion and scale variance.

From this fused representation F, we branch into a decoupled detection head with two parallel branches—a design inspired by the YOLOX framework [53]—to disentangle classification and localization tasks. The classification head applies two 3×3 convolutions followed by a 1×1 kernel to predict per-cell objectness scores p^{obj} and class logits $\{p^1, \ldots, p^C\}$ over $C = 7$ categories:

$$O \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times K(1+C)} \tag{6}$$

Simultaneously, the regression head, which has an identical structure, outputs bounding-box offsets (t_x, t_y, t_w, t_h) for each prediction point:

$$B \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times 4K} \tag{7}$$

where K denotes the number of candidate prediction points at each spatial location. The offsets t_x, t_y are passed through a sigmoid function and combined with cell indices i, j to obtain absolute center coordinates, while t_w, t_h are transformed to estimate the width and height of the predicted bounding box. Decoupling classification and regression mitigates conflicting gradient signals and improves convergence as well as small-object localization accuracy.

To train MicroPPE-Guard end-to-end, we minimize a composite loss:

$$\mathscr{L} = \lambda_{\text{obj}} BCE\left(p^{\text{obj}}, \widehat{y^{\text{obj}}}\right) + \lambda_{\text{cls}} CE(p^c, \widehat{y^c}) + \lambda_{\text{reg}} CIoU(b, \widehat{b}) \tag{8}$$

where *BCE* is binary cross-entropy on objectness, *CE* is multi-class cross-entropy on class labels, and *CIoU* is the Complete-IoU loss that jointly penalizes overlap, center-distance, and aspect-ratio discrepancy. We set $\lambda_{\text{obj}} = 1.0, \lambda_{\text{cls}} = 1.0$ and $\lambda_{\text{reg}} = 2.5$ to strike a suitable balance between detection confidence and bounding-box precision.

Finally, recognizing that video-based monitoring demands temporal stability and spatial validity, we append the Temporal–Spatial Compliance Module (TSCM). First, detections are matched frame-to-frame using the Hungarian algorithm with IoU-based cost. For each matched detection track, the confidence score is smoothed using an exponential moving average:

$$\widehat{s}_t = \alpha s_t + (1 - \alpha)\widehat{s}_{t-1}, \alpha = 0.6 \tag{9}$$

where s_t is the current confidence score and $\widehat{s}_{t-1}$ is the previous smoothed score. This reduces flickering under camera movement, motion blur, and transient occlusions. Second, we enforce that each detected PPE item overlaps sufficiently with a person's bounding box by computing the overlap ratio:

$$\rho = \frac{\text{Area}\left(b_{ppe} \cap b_{person}\right)}{\text{Area}\left(b_{ppe}\right)} \tag{10}$$

A PPE detection is discarded if:

$$\rho < \tau, \tau = 0.5 \tag{11}$$

This geometric check ensures, for example, that a glove lying on a toolbox is not mistakenly interpreted as being worn by a worker, thereby reducing false positives in busy work zones.

6.2. Experimental setup

To evaluate the effectiveness of the MicroPPE-Guard model, we conducted extensive experiments on three distinct datasets: the proposed 25K, a high-resolution PPE dataset curated in this study to capture diverse, real-world construction scenarios; UAVOD-10, a publicly available benchmark of 844 annotated images used to assess generalization to aerial and wide-area monitoring conditions [29]; and NewPPE25, the PPE dataset introduced by Huang and Cheng [54], which serves as an external benchmark to validate robustness across heterogeneous data sources. The composition of the proposed 25K dataset, its seven PPE classes, and the source-grouped 70/20/10 train/validation/test split are detailed in Sections 3.1–3.2 and Table 2.

To ensure fair comparison and improve reproducibility, all models were trained and evaluated on the same fixed train/validation/test split manifest—generated once using the source-grouped 70/20/10 protocol of Section 3.2, with no per-run random seed—and under an identical hardware environment (a single NVIDIA RTX 3090 Ti), input-resolution protocol, confidence threshold (0.25), NMS IoU threshold (0.50), and evaluation metrics (mAP_{50} and F1 score), all listed in Table 4. Inference latency refers solely to model-forward time per frame; camera decoding, video streaming, display rendering, and alert-interface overhead were excluded. The main experimental configuration is summarized in Table 4.

For the non-YOLO baselines, all models were evaluated as complete object-detection pipelines rather than as standalone classification backbones. Specifically, ResNet, VGG, EfficientNet, MobileNetV3, and Swin Transformer were implemented within Faster R-CNN + ResNet-50-FPN, SSD + VGG16, EfficientDet-D0, SSD-MobileNetV3, and a Swin-T-based detector, respectively. For the YOLO-family baselines, we applied the same training split, input size, evaluation metrics, confidence threshold, and NMS threshold, ensuring that performance differences primarily reflect model design rather than inconsistent experimental settings.

MicroPPE-Guard uses a CSPDarknet-style backbone initialized with ImageNet-pretrained weights and is trained end-to-end for 300 epochs. The optimizer, batch size, number of epochs, momentum, and weight decay are reported in Table 4; the initial learning rate, learning-rate schedule, warm-up configuration, and data-augmentation settings followed the default configuration of the training implementation and were not manually tuned. Training requires approximately 48 h to converge on a single RTX 3090 Ti. Within the complete 1920×1080 single-stream monitoring pipeline (batch size 1), MicroPPE-Guard processes frames at approximately 35 FPS, measured under FP32 (single-precision) inference. Ablation studies were conducted on the proposed 25K dataset, using mAP_{50} and F1 score, by disabling or replacing individual components, namely the enhanced FPN, SE fusion block, decoupled detection head, TSCM temporal smoothing, and spatial compliance validation.

7. Experimental results

This section reports the quantitative and qualitative performance of MicroPPE-Guard under the experimental protocol described in Section 3.4. The evaluation covers three datasets: the in-house proposed 25K dataset, the UAVOD-10 generic small-object benchmark, and the external NewPPE25 PPE dataset. A synthetic SmallPPE subset—derived from the proposed 25K dataset to emphasize extremely small objects—is additionally evaluated in Table 8; it is a subset of the 25K data rather than an independent external dataset. The comparison spans YOLO-family baselines, lightweight detectors, two-stage detectors, and Transformer-based detection pipelines. Table 5 summarizes the overall

Table 4
Experimental configuration for reproducible and fair comparison.

Configuration	Setting
Training datasets	Proposed 25K, UAVOD-10, and NewPPE25
Dataset split	70% training, 20% validation, and 10% testing
Hardware	Single NVIDIA RTX 3090 Ti
Training epochs	300
Batch size	16
Optimizer	Stochastic Gradient Descent
Momentum	0.937
Weight decay	5×10^{-4}
Confidence threshold	0.25
NMS IoU threshold	0.50

Table 5
Comparative detection performance, throughput, and memory usage.

Dataset	Model	mAP_{50}	F1	Inference time (ms)	Memory (MB)
Proposed 25K	MicroPPE-Guard	0.924	0.913	15	171
	YOLOv12n	0.891	0.883	13	132
	YOLOv11n	0.893	0.871	14	143
	YOLO-MS	0.889	0.834	18	162
	Swin-T Detector	0.883	0.810	40	352
	Faster R-CNN + ResNet-50-FPN	0.801	0.779	35	300
	EfficientDet-D0	0.790	0.760	27	215
	SSD + VGG16	0.780	0.750	32	250
	SSD-MobileNetV3	0.710	0.720	18	140
UAVOD-10	MicroPPE-Guard	0.650	0.600	16	**125**
	YOLOv12n	0.550	0.500	23	175
	YOLOv11n	0.540	0.480	21	155
	YOLO-MS	0.580	0.530	16	105
	Swin-T Detector	0.510	0.470	41	405
	Faster R-CNN + ResNet-50-FPN	0.500	0.450	36	305
	EfficientDet-D0	0.490	0.440	27	215
	SSD + VGG16	0.480	0.430	33	215
	SSD-MobileNetV3	0.310	0.290	19	140
NewPPE25	MicroPPE-Guard	0.780	0.760	**17**	**128**
	YOLOv12n	0.770	0.730	12	178
	YOLOv11n	0.750	0.690	14	158
	YOLO-MS	0.760	0.730	22	208
	Swin-T Detector	0.690	0.640	42	408
	Faster R-CNN + ResNet-50-FPN	0.670	0.640	37	308
	EfficientDet-D0	0.740	0.720	17	128
	SSD + VGG16	0.730	0.710	12	178
	SSD-MobileNetV3	0.310	0.290	19	140

detection accuracy, inference latency, and memory usage across all evaluated models.

MicroPPE-Guard's superior performance on the proposed 25K dataset stems primarily from its tailored multi-scale feature extraction and fusion strategy, which explicitly targets the varied object sizes and occlusion patterns encountered in real-world PPE monitoring. By pairing a Darknet-style backbone with an enhanced Feature Pyramid Network (FPN), the model preserves high-resolution detail in the early layers—essential for localizing small items such as gloves or safety glasses—while integrating deeper semantic cues in later layers to disambiguate PPE from complex backgrounds. Leaner YOLO variants (including YOLOv12n and YOLOv11n), by contrast, trade some of this fine-grained spatial information for speed, yielding a modest drop in both mAP_{50} and F1. Heavier detector configurations, such as the Swin-T Detector, Faster R-CNN + ResNet-50-FPN, SSD + VGG16, and EfficientDet-D0, do capture rich contextual features, but their inference pipelines incur greater latency and memory overhead; moreover, they often underperform on tiny objects because their deeper layers tend to smooth out fine details. MicroPPE-Guard's channel-wise attention fusion block (Squeeze-and-Excitation) further amplifies the feature channels most relevant to each PPE class, which may contribute to improved precision on headgear and vests while down-weighting irrelevant background textures. Combined with extensive online augmentation (mosaic, mixup, hue/brightness jitter, blurring, and noise), this attention mechanism yields representations that generalize well to the varied lighting and pose conditions of the proposed 25K dataset. The UAVOD-10 results should be interpreted as evidence of generic small-object transferability rather than direct PPE-compliance validation, since UAVOD-10 contains non-PPE aerial objects; the NewPPE25 results, in turn, indicate that the model remains competitive on an external PPE data source.

To examine the detection behavior of MicroPPE-Guard across individual PPE-related categories, we report per-class AP_{50}, $AP_{50:95}$ and recall on the proposed 25K test set. This analysis is particularly important because several PPE categories—notably safety glasses, gloves, and safety boots—are often small, partially occluded, or visually ambiguous in construction-site imagery.

As shown in Table 6, MicroPPE-Guard attains higher AP_{50} on larger, visually distinctive categories such as persons, reflective vests, and helmets, whereas smaller PPE categories such as safety glasses, gloves, and safety boots remain more challenging. The lower $AP_{50:95}$ values for these small categories indicate that localization precision degrades under stricter IoU thresholds, particularly when PPE occupies only a small portion of the image or is partially occluded by body parts, tools, or surrounding construction materials. Even so, the model maintains relatively balanced recall across all seven categories (ranging from 0.854 to 0.951), suggesting that the enhanced FPN and SE-based fusion help preserve small-object cues without sacrificing detection performance on larger safety-related classes. To complement these quantitative results, Fig. 5 presents representative qualitative examples under challenging PPE-monitoring conditions, spanning indoor and outdoor construction scenes, long-distance CCTV-style views, dense worker layouts, partial occlusion, and small PPE instances such as gloves, boots, and safety-related head regions.

As shown in Fig. 5, MicroPPE-Guard delivers consistent localization across diverse construction-site conditions, detecting larger safety-related targets such as workers and safety vests while retaining sensitivity to smaller PPE items such as gloves and boots. These qualitative results reinforce the per-class analysis, indicating that the proposed multi-scale fusion strategy and channel-attention design remain effective even when PPE items are small, partially blocked, or embedded in visually complex backgrounds.

To quantify the contribution of each component in the proposed framework, we conducted a module-wise ablation study on the proposed 25K dataset. Starting from a baseline YOLO-style detector, we added the proposed modules one at a time—the enhanced FPN, the SE fusion block, the decoupled detection head, and the Temporal–Spatial Compliance Module (TSCM). The results are reported in Table 7.

The enhanced FPN yields the largest single improvement, primarily because it strengthens multi-scale feature representation for small PPE targets. This matters most in construction-site imagery, where objects such as gloves, boots, and safety glasses may occupy only a small portion of the frame. By improving the flow of scale-sensitive information across feature levels, the enhanced FPN helps the model retain fine spatial details that standard detection pipelines tend to lose.

The SE fusion block adds a further gain by steering the model toward more informative feature channels. This is valuable in complex construction environments, where PPE items may be visually mixed with body parts, tools, machinery, or background materials. By emphasizing discriminative channels and suppressing less useful responses, SE fusion helps separate PPE regions from cluttered or visually similar surroundings. The decoupled detection head contributes a more targeted gain: by splitting classification and localization into separate branches, it reduces task conflict and lets the model strike a better balance between category prediction and bounding-box accuracy.

Although the Temporal–Spatial Compliance Module (TSCM) yields a smaller increase in mAP_{50}, its contribution is more evident in the F1 score. This indicates that TSCM is not mainly designed to improve isolated frame-level detection; rather, its value lies in reducing unstable predictions, duplicate detections, and geometrically inconsistent PPE–person associations across frames. In practical safety monitoring, such stability is essential, because the system must deliver reliable compliance judgments rather than outputs that fluctuate from frame to frame. Overall, the ablation confirms that each proposed component addresses a distinct limitation of the baseline, and these gains are obtained while maintaining real-time inference efficiency.

On UAVOD-10, MicroPPE-Guard is evaluated as a general small-object detection architecture rather than a PPE-specific compliance

Table 6
Per-class detection performance of MicroPPE-Guard on the proposed 25K test set.

Class	Annotation label	AP_{50}	$AP_{50:95}$	Recall	Main difficulty
Person	person	0.958	0.742	0.951	Large object; affected by crowding and occlusion
Head	head	0.932	0.658	0.910	Small region; affected by helmet/head overlap
Helmet	hat	0.945	0.690	0.928	Confusion with uncovered head or background objects
Glove	glove	0.899	0.532	0.872	Tiny object; frequent hand motion and occlusion
Safety glasses	glasses	0.881	0.501	0.854	Very small object; weak visual texture
Safety boots	boot	0.902	0.552	0.868	Lower-body occlusion and scale variation
Reflective vest	vest	0.951	0.712	0.936	Shape deformation and partial visibility
Mean	—	0.924	0.627	0.903	—



Fig. 5. Qualitative detection examples of MicroPPE-Guard under challenging PPE-monitoring conditions. (a) Indoor industrial scene with small gloves, boots, head, vest, and worker localization; (b) Top-view construction scenario with multiple workers, partial occlusion, and small PPE instances; (c) Long-distance CCTV-style scene with workers on elevated platforms and small PPE objects; (d) Outdoor construction scene with dense workers, boots, vests, and head/helmet-related detections under cluttered background conditions.

Table 7
Ablation study of MicroPPE-Guard components on the proposed 25K dataset.

Variant	Enhanced FPN	SE fusion	Decoupled head	TSCM	mAP_{50}	F1	Inference time (ms)	Memory (MB)
YOLO-style detector	×	×	×	×	0.886	0.858	13	142
+ Enhanced FPN	✓	×	×	×	0.904	0.879	14	156
+ Enhanced FPN + SE Fusion	✓	✓	×	×	0.912	0.889	15	165
+ Enhanced FPN + SE Fusion + Decoupled Head	✓	✓	✓	×	0.920	0.904	15	169
Full MicroPPE-Guard	✓	✓	✓	✓	0.924	0.913	15	171

Table 8
Comparative detection performance, throughput, and memory usage on the SmallPPE dataset.

Dataset	Model	mAP_{50}	F1	Inference time (ms)	Memory (MB)
Proposed SmallPPE	MicroPPE-Guard	0.72	0.70	26	135
	YOLOv12n	0.67	0.65	12	91
	YOLOv11n	0.66	0.61	14	125
	YOLO-MS	0.71	0.69	18	105
	Swin-T Detector	0.68	0.65	44	405
	Faster R-CNN + ResNet-50-FPN	0.48	0.43	32	305
	EfficientDet-D0	0.42	0.41	37	215
	SSD + VGG16	0.43	0.42	45	215
	SSD-MobileNetV3	0.38	0.35	30	140

framework. Because UAVOD-10 comprises aerial object categories unrelated to PPE, the detection head was retrained to match the UAVOD-10 label space while the proposed multi-scale feature refinement and attention-based fusion modules were retained. The results show that MicroPPE-Guard remains effective for aerial small-object detection, where targets are often distant, weakly represented, and set against complex backgrounds.

Relative to YOLO-based baselines, the enhanced FPN and SE-based fusion modules better preserve the fine-grained spatial cues that UAV imagery tends to weaken through scale variation, long viewing distances, and background clutter. This suggests that the proposed architecture has transfer potential beyond construction-site PPE monitoring and may benefit broader small-object detection tasks. The TSCM-based PPE–person compliance reasoning, however, remains application-specific, as it is designed for safety-monitoring scenarios in which the relationship between workers and PPE items must be explicitly evaluated.

On the NewPPE25 dataset, whose data are drawn from unseen indoor and outdoor sites, MicroPPE-Guard's extensive augmentation strategies and balanced loss function—which carefully weights objectness, classification, and localization—ensure robust feature learning that transfers across domains without overfitting. The decoupled classification/regression heads reduce conflicting gradients during training, accelerating convergence and preserving small-object sensitivity. By comparison, YOLOv12n's more compact head design yields a slightly lower raw mAP_{50} (0.770 vs. 0.780) and lower F1, indicating a less balanced precision–recall trade-off. Heavier backbones again falter, owing to slower convergence on novel data and reduced pixel-level detail in their feature maps; this manifests as lower overall accuracy—for example, the Swin-T Detector's 0.690 mAP_{50} and 0.640 F1. MicroPPE-Guard's ability to sustain a high F1 (0.760) at an inference time of 17 ms and 128 MB of GPU memory stems from its architectural synergy: multi-scale fusion for small-object fidelity, channel attention for context discrimination, decoupled heads for stable training, and compliance validation for post-processing consistency. This helps explain why the model consistently matches or outperforms competing detectors across all evaluated datasets and benchmark settings.

On the SmallPPE dataset—purpose-built to emphasize tiny, hard-to-detect items such as gloves and safety glasses—MicroPPE-Guard again ranks among the top performers, achieving a mAP_{50} of 0.72 and an F1-score of 0.70, albeit at a slightly higher inference cost (26 ms per 1080p frame, 135 MB). This edge arises from the same multi-scale fusion and channel-attention design that excels at preserving minute pixel detail: the enhanced FPN keeps the early convolutional layers responsible for small-object localization prominent, while the Squeeze-and-Excitation module adaptively highlights the feature channels most indicative of tiny PPE regions. YOLOv12n and YOLOv11n, though faster (12 ms/91 MB and 14 ms/125 MB, respectively), trade the necessary fine-grained spatial resolution for speed and compactness, resulting in lower mAP_{50} (0.67 and 0.66) and F1 (0.65 and 0.61). YOLO-MS performs comparably to MicroPPE-Guard ($mAP_{50} = 0.71$, F1 = 0.69) by adding small-object anchors and refined training hyperparameters, trailing by only 0.01 in mAP_{50}; this narrow margin reflects comparable performance rather than a claim of clear superiority or statistical significance. The transformer-based Swin-T Detector reaches a competitive mAP_{50} of 0.68 and F1 of 0.65, but at a prohibitive 44 ms and 405 MB; its deep attention layers dilute pixel-level granularity and inflate resource demands, making it less practical for real-time monitoring. The traditional detector configurations—Faster R-CNN + ResNet-50-FPN (mAP_{50} = 0.48, F1 = 0.43), EfficientDet-D0 (0.42, 0.41), and SSD + VGG16 (0.43, 0.42)—struggle to preserve tiny PPE cues because their receptive fields are already too coarse by the time features reach the final prediction layers, yielding both dramatically lower accuracy and moderate inference latencies (32–45 ms, 215–305 MB). SSD-MobileNetV3, though optimized for efficient deployment, falls short ($mAP_{50} = 0.38$, F1 = 0.35) because its aggressive depthwise separable convolutions compress feature maps and erase the subtle textures that delineate gloves or safety glasses. In sum, on SmallPPE the proposed MicroPPE-Guard strikes an optimal balance: multi-scale fusion and channel attention preserve high-resolution detail to maximize small-object recall and precision while keeping runtime within acceptable real-time bounds, whereas both ultra-lightweight models and very heavy detectors either lose crucial detail or incur untenable latency and memory costs.

Because TSCM is intended to improve temporal stability and compliance consistency rather than only static detection accuracy, we conducted an additional evaluation on video-derived samples from the proposed 25K dataset. Table 9 compares the detector-only setting against EMA smoothing, spatial compliance checking, and the full TSCM configuration.

As reported in Table 9, TSCM does not substantially increase mAP_{50}, but its main contribution is to make the detections more stable across video frames. In real surveillance footage, predictions can fluctuate from frame to frame, particularly when PPE items are small, partially blocked, or affected by motion. TSCM mitigates this by smoothing detection confidence over time and verifying that each detected PPE item is spatially associated with a worker.

Specifically, EMA smoothing reduces flickering predictions by making confidence scores more consistent across neighboring frames, while the spatial compliance check improves reliability by filtering out PPE detections that are not properly associated with a detected person, thereby avoiding false alerts triggered by isolated or spurious PPE objects in the background. Combined, the two components yield the full TSCM setting, which offers the best balance among detection performance, temporal stability, and compliance judgment while adding almost no extra inference cost.

To assess whether TSCM remains reliable under varying video conditions, we also conducted a sensitivity analysis. It examined several practical factors: the spatial overlap threshold, frame-rate changes, motion blur, and the sudden appearance or disappearance of objects. The default configuration used an overlap threshold of $\tau = 0.5$ and video input at 30 FPS. The results are summarized in Table 10.

As shown in Table 10, the default threshold $\tau = 0.5$ provides the best balance between detection accuracy and compliance consistency. A lower threshold $\tau = 0.3$ retains more uncertain PPE detections but increases false alerts, while a stricter threshold $\tau = 0.7$ reduces false alerts at the cost of a lower F1. Reducing the frame rate from 30 FPS to 15 FPS increases flicker, because fewer adjacent frames are available for temporal smoothing. Motion blur and the sudden appearance or disappearance of objects also reduce stability, yet TSCM still maintains lower false-alert and flicker rates than detector-only inference. Overall, TSCM remains effective under realistic video disturbances, and the default threshold stays stable across common surveillance conditions.

8. Conclusion and discussion

8.1. Conclusion

In this work, we introduced MicroPPE-Guard, a lightweight yet high-precision, real-time-capable PPE detection framework. By integrating a multi-scale Darknet-style backbone with an enhanced Feature Pyramid Network (FPN), channel-wise attention fusion (Squeeze-and-Excitation), decoupled classification and regression heads, and a Temporal–Spatial Compliance Module (TSCM), MicroPPE-Guard preserves the fine spatial detail needed to identify small items such as gloves and safety glasses while exploiting the deeper semantic cues needed to distinguish PPE from complex backgrounds. On our in-house Proposed 25K dataset, the framework achieved $mAP_{50} = 0.924$ and F1 = 0.913 at approximately 35 FPS for the end-to-end 1920 × 1080 single-stream pipeline (model-forward time 15 ms) with a 171 MB GPU footprint. On the UAVOD-10 aerial benchmark, which contains non-PPE small objects that appear as tiny, distant instances, MicroPPE-Guard attained $mAP_{50} = 0.650$ and F1 = 0.600 at 30 FPS (model-forward time 16 ms; 125 MB), outperforming the evaluated YOLO-family baselines on this generic small-object task; this result demonstrates transferability to generic small-object detection and does not, by itself, constitute evidence of PPE-

Table 10
Sensitivity analysis of TSCM under threshold and video-condition variations.

Setting	Configuration	mAP_{50}	F1	Flicker rate	False alert rate
Spatial threshold	$\tau = 0.3$	0.923	0.908	0.101	0.092
Spatial threshold	$\tau = 0.5$	0.924	0.913	0.097	0.078
Spatial threshold	$\tau = 0.7$	0.919	0.904	0.096	0.071
Frame rate	15 FPS	0.920	0.906	0.124	0.084
Frame rate	30 FPS	0.924	0.913	0.097	0.078
Degradation	Motion blur	0.914	0.899	0.138	0.096
Degradation	Sudden appearance/ disappearance	0.918	0.903	0.146	0.089

Table 9
Effect of TSCM on temporal stability and compliance consistency.

Variant	EMA smoothing	Spatial compliance check	mAP_{50}	F1	Flicker rate	False alert rate	Inference time (ms)
Detector only	×	×	0.920	0.904	0.184	0.132	15
+ EMA smoothing	✓	×	0.921	0.908	0.121	0.116	15
+ Spatial compliance check	×	✓	0.922	0.909	0.168	0.091	15
Full TSCM	✓	✓	0.924	0.913	0.097	0.078	15

compliance generalization. Under domain shift on NewPPE25 (unseen indoor and outdoor sites), it maintained strong performance (mAP_{50} = 0.780, F1 = 0.760) at 30 FPS (model-forward time 16 ms; 128 MB), demonstrating robustness across diverse environments.

Across the three main benchmarks—the proposed 25K, UAVOD-10, and NewPPE25—MicroPPE-Guard achieved a competitive balance among detection accuracy, inference speed, and memory efficiency for small PPE monitoring.

8.2. Discussion

By combining multi-scale feature extraction, channel-wise attention, and decoupled detection heads, MicroPPE-Guard identifies both large and small PPE items precisely without incurring excessive computational cost. Across our benchmarks, this architecture outperformed both lighter YOLO variants, which trade detail for speed, and heavier backbones, which sacrifice real-time throughput, striking an optimal balance between accuracy (e.g., mAP_{50} = 0.924 on proposed 25K) and resource usage (≈ 35 fps with ~ 171 MB); on an RTX 3090 Ti, it processed a single 1920 × 1080 monitoring stream at approximately 35 FPS. The enhanced FPN keeps tiny objects such as gloves and safety glasses detectable even when they occupy only a few pixels, while SE fusion adaptively emphasizes the most informative channels, reducing false positives in cluttered scenes.

Several technical limitations remain, the most notable concerning the detection of small PPE items. Compared with workers, helmets, and safety vests, objects such as gloves and safety glasses are far harder to recognize because they typically appear very small in the frame. In some surveillance images they occupy only a few pixels and may lack a clear boundary altogether, and motion blur, partial occlusion, and background clutter compound the difficulty. A glove, for example, can easily blend into a worker's hand, a tool, or nearby construction material, and safety glasses may be barely visible when the worker is far from the camera or facing away from the lens. These cases show that strong overall detection performance does not always guarantee reliable recognition for every PPE category.

Deployment is a second issue that remains only partially addressed. In this study, MicroPPE-Guard proved lightweight under controlled GPU-based testing, but this does not necessarily mean that the system is ready for direct deployment on embedded devices. The reported FPS mainly reflects single-GPU inference; in real construction-site environments, the model may face additional constraints such as restricted memory, heat-related performance drops, camera-stream latency, and hardware-specific optimization difficulties. These factors can degrade real-time performance even when the model appears efficient in laboratory evaluation.

Future work will therefore proceed along several practical directions. The first is to test the model on embedded platforms, particularly Jetson-series hardware, so that its real-time performance can be assessed under more realistic deployment conditions. We will also expand the dataset with additional CCTV footage from actual construction sites, giving greater attention to difficult cases such as extremely small PPE items, blurred objects, heavy occlusion, and complex worker interactions. In addition, TSCM will be compared with stronger tracking-by-detection methods, including ByteTrack, DeepSORT, and StrongSORT, to determine whether more advanced tracking strategies can improve temporal stability and PPE–person association. Finally, future versions may incorporate selective explanation modules for medium-confidence or safety–critical detections, particularly where human confirmation is required before a compliance alert is sent.

9. Data availability

Due to privacy restrictions, construction-site confidentiality, and limitations on third-party image sources, the full raw image dataset cannot be released publicly at this stage. However, the data will be shared privately upon request for academic research purposes, provided that requesters sign a confidentiality agreement.

10. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

In preparing this paper, the authors used an AI writing assistant to check the document's spelling and grammar and to improve the language and clarity of the draft. After using the tool, the authors reviewed the document, made necessary changes, and took responsibility for its content.

CRediT authorship contribution statement

Xuan Le: Writing – original draft, Software, Methodology, Data curation. **Phu Pham:** Writing – review & editing, Validation, Methodology, Investigation. **Kien-T. Le:** Validation, Formal analysis, Data curation. **Luan Bui:** Validation, Software, Data curation. **Bao Huynh:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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